

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the
Interstate Transmission System

Docket No. RM26-4-000

COMMENTS OF SHELL ENERGY NORTH AMERICA (US), L.P.

Shell Energy North America (US) LP (“Shell Energy”) appreciates the opportunity to provide comments on the ANOPR in the above-referenced docket.¹

I. EXECUTIVE SUMMARY

Shell Energy supports the objectives and direction outlined in the ANOPR, which seeks to establish a transparent, equitable, and efficient process for the timely and organized interconnection of large loads to the transmission system, without compromising grid reliability.

Shell Energy’s comments respond to the proposed policy principles on effective reforms to expedite interconnecting large loads, and not on the federal and state jurisdictional roles described in the ANOPR. Competitive market participants value regulatory certainty and legally durable processes as a prerequisite to major capital investments; the market will carefully follow the jurisdictional roles discussion as it unfolds.

FERC has established its own principles and set of legally durable criteria on how to achieve its mission to facilitate economically efficient markets, and criteria to determine just and reasonable rates. FERC precedent prioritizes market transparency, competition to support consumer affordability, and maintaining regulatory certainty by preserving the rights of existing market participants. Incremental progress and innovation can proceed regionally concurrently with the ANOPR; indeed, the implementation experience integrating large loads in individual regions to-date provides valuable information about the tradeoffs FERC must consider in this rulemaking.

Reforms that increase accuracy of the load projections used in the FERC-jurisdictional capacity markets should be prioritized. It is crucial for both regulators and market participants to identify the true scale of load growth, because their decisions—such as which design solutions regulators choose and how much investment risk market participants assume—will vary greatly

¹ Shell Energy is one of the largest traders of electricity, natural gas, and environmental products in North America and also actively participates in wholesale electric markets across North America. In addition, Shell Energy has affiliates that invest in and develop clean energy resources, supply retail power to customers, and serve large end users.

depending on whether the increase is 15GW or 50GW over the next five years. The current forward markets are equipped to meet the needs of increased demand when forecasts used in RTO/ISO administered competitive markets are credible and timely, or where the risks to load forecast accuracy are transparent. Even if other proposed reforms require longer processes to complete because of the challenging issues under consideration, FERC, in accordance with its jurisdiction over wholesale markets, is authorized to set regulations stipulating that capacity markets should consider those large loads that have satisfied the required study deposits and readiness criteria, and not include large loads that have not paid deposits or demonstrated readiness. As it does not require complex engineering or policy tradeoffs, it can be implemented quickly. This skin in the game from developers will increase transparency, which in turn will enable forward markets to reflect the value of the actual load and trigger immediate improvement in the commercial response to the more realistic projected demand.

It is essential that, during the rulemaking process, careful consideration is given to prevent undue discrimination against new or existing load and generation resources, and to ensure that wholesale rates remain just and reasonable. To facilitate the transition through any rulemaking processes, we provide key considerations and clarifying guidance to remove potential misinterpretation of the ANOPR principles and move toward fast implementation.

Key considerations:

- First and highest priority should be for improving accuracy of load projections used in current competitive markets.
- Ensure the new rules do not disrupt market price signals and misalign with established market principles.
- Allow market flexibility by avoiding mandates or barriers so competitive markets can innovate to ensure reliability cost-effectively.
- Ensure restricted access designed to speed up the interconnection process in the interim doesn't preclude future requests for full access utilizing normal processes.
- Recognize regional differences in wholesale market designs and grid needs.
- Protect existing generation interconnection queue resources from negative impacts of new processes.
- The rulemaking process should not impede the ongoing development of regional programs, tariffs, and procedures that support the objectives outlined in the ANOPR.

Shell Energy proposes the following guidance where the ANOPR initial principles were silent on key implementation details:

- 1) The principles should not be requirements for large loads to interconnect under permanent operational restrictions; instead, should be interpreted as options that can expedite initial

interconnections and allow for interim operations while full studies and transmission construction activities are completed if large loads need full grid access. Shell Energy supports efforts to standardize processes across regions and improve speed to market for large loads, but permanent restrictions on large load operations or removing the option to proceed through study for full participation does not facilitate access to competitive market solutions in the long run. (Principles 3, 5, 6, 7, 10, 11, 13, 14)

- 2) Where possible, the ANOPR should build upon existing market processes, definitions, and tools in implementation, instead of creating new definitions and tools only applicable to qualifying large loads. Shell Energy believes that using existing market tools speeds implementation, since establishing if the characteristics of large loads or if the characteristics of hybrid/co-located large loads with generating resources require distinct treatment (relative to existing load and generation resources) is a time-consuming prerequisite for preferential or discriminatory treatment. (Principles 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14)
- 3) The Talen/PJM co-location docket and combined Technical Conference² established that the decisions are technically complex and nuanced on where it is appropriate to use net injection/withdrawal in study practices versus gross injection and withdrawal separately. As we note below when responding to individual principles, the RTO/ISOs and NERC should weigh in if the reliability risks typically assessed during interconnection studies (transient stability, voltage, harmonics) can be done in the expedited study timeframe, and if those risks to the BPS are mitigated by system protection facilities or system protection schemes. Additionally, large load end-users need to weigh in if such an operating model fits their power needs, or if it is prohibitive to large load operations to be hybrid/co-located behind system protection facilities or system protection schemes with strict limits on withdrawals or injection to the grid around the clock. (Principles 3, 5, 6, 7, 10, 11, 13, 14)
- 4) Even where net injection/withdrawal is the conceptually appropriate quantity, there are a variety of metrics used in current supply/demand accounting frameworks, transmission planning, resource adequacy frameworks, and energy market operations. Shell Energy requests the rulemaking harmonize the calculation of net injection/withdrawal with existing resource adequacy accreditation rules, which vary by region. These accreditation methodologies may include UCAP, EFORD, ELCC, marginal ELCC, or exceedance, measured across a range of assessment periods, such as during annual peak load hour only, across monthly/seasonal load peaks, and in long-term risk-based scenarios as required by Order No. 1920. The ANOPR did not specify if other physical parameters, such as ramp rate, frequency response, load following capability, economic minimum, and economic maximum must be reasonably well matched with expected load parameters to ensure it is feasible to operate in a hybrid manner. When paired with system protection facilities or system protection schemes that will prevent or penalize injection or withdrawal outside

² EL25-49-000, EL25-20-000 Consolidated, and AD24-11-000 Large Loads Co-Located at Generating Facilities

narrow tolerance bands, Shell Energy supports matching additional operating criteria (beyond capacity accreditation) to feasibly operate hybrid/co-located generation and load. (Principles 3, 5, 6, 7, 10, 11, 13, 14)

II. COMMENTS

A. Applicability of rules

Principle 1: Limiting this rulemaking to transmission interconnections instead of distribution system interconnections could help mitigate the debate on more complicated and time consuming federal jurisdiction issues.

Creating a standardized and transparent load interconnection queue for transmission connected large loads would capture a significant portion of this new demand, facilitating the ability of all market participants to perform effective risk analysis and reflect the value of these loads in forward pricing, enabling them to site investment appropriately.

Principle 2: There is already regional variation in how a threshold has been designed- e.g. ERCOT has set a 75 MW threshold, SPP created MW thresholds based on the transmission system voltage where the large load is interconnecting, and other existing definitions such as Behind the Meter size caps may create a just and reasonable threshold. Existing large load customers that entered a load interconnection queue before thresholds were established should be included in future reporting for transparency.

Shell Energy agrees that a minimum size threshold is necessary. It should be high enough to minimize the administrative burden to RTO/ISOs of managing multiple study processes, and high enough to minimize impacts to traditional, non-large load customers requesting service. At the same time, the minimum thresholds should be low enough that, when combined with principle 4 (study deposits and readiness criteria), load forecasts accurately capture the scale of the total load additions that could impact resource adequacy and wholesale market prices. Regional stakeholder processes are well situated to establish the exact level based on reliability risks, the demand for load interconnection in the region, and existing RTO/ISO processes.

B. Study Practices and Physical or Technical specifications

Principles 3 and 5: As background, existing interconnection studies determine both a) if there is sufficient transmission at peak hours to simultaneously and feasibly bring energy from generation to load, as well as b) other transmission stability elements, such as the fault contributions of new power supply, how the new supply affects the voltage profile of the system, etc. Shell Energy supports analyzing the net impact of hybrid/co-located load with generation to allow for more efficient siting, faster study processing and shorter transmission infrastructure construction timelines. In regions where generators must own firm transmission rights to be eligible to sell resource adequacy, this rulemaking should clarify that generation resources are

eligible to sell capacity or resource adequacy products to co-located load without requiring firm transmission rights.

The transmission stability elements of generator and load interconnection studies can also trigger required network upgrades. This rulemaking should clarify that net injection/withdrawal applies to both the resource adequacy transmission rights and steady state transmission stability elements of a full interconnection study. Typically, steady state transmission planning studies are based on the full characteristics of the generation/load and not based on the net injection/withdrawal. In regard to transient voltage and stability control studies, more input from the RTO/ISOs and NERC is needed to determine if such studies can be done on an expedited and reliable manner outside of the cluster process, or what type of system protection facilities or schemes would mitigate the need to study transient voltage & stability control risks in the area when studying at net injection/withdrawal level.

Shell Energy appreciates the use of net injection/withdrawal in interconnection studies of hybrid/co-located load and generation to expedite interconnection, to the extent it can be done while maintaining grid stability and reliability. However, the requirement to net and install special protective schemes should not be mandated as a permanent option precluding the entities from requesting to connect at gross level unless agreeable to the hybrid/co-located load and generation customer(s). This could be implemented as a potential interim option, giving hybrid/ co-located load and generation resources the right to request future study and full market participation as distinct load and generation if that better suits their long-run operational needs.

Principle 4: At the time of entry into load queue, Shell Energy supports study deposits, readiness requirements including application fees, and withdrawal penalties for load interconnection requests similar to existing generator interconnections studies and application fees. These measures deter speculative projects, discipline the size of requests, and increase confidence when including large loads in long-range planning forecasts. Readiness criteria should also include attestations about duplicative requests, proof of site control for both the facility site and tie-line, and application for material permits. Posting of application fees, study fees, cost in aid of construction and/or early \$/MW financial securities should be eligible for refund following completion of a harm assessment showing no material impact on equally or lower queued customers if the large load project withdraws its application. These application fees or security deposits, submitted at the time of large load application, could go toward network upgrades identified in service agreement after study. If these deposits exceed the total cost of study and assigned network upgrades, these excess deposits could be refunded after the load has been placed in service, in proportion to how much of the load materialized relative to full requested MWs of requested load interconnection.

Shell Energy views this as the most important principle proposed in the ANOPR. This reform, with proper transparency reporting, significantly improves the credibility of the load forecasts. These deposits should be carefully structured - with early posting of meaningful \$/MW

security, acceptable types of security accepted, early collection of securities, refunds only up to realized load levels, etc. - to ensure that applicants are genuinely committed. This approach encourages the reduction of speculative or duplicate applications and enhances the accuracy of load projections. Increased scrutiny and transparency will strengthen market confidence in these projections, allowing forward markets to more accurately reflect actual loads and enabling participants to make informed investment decisions. These fees may need to be adjusted in the future for inflation, so it would be less onerous on the Commission to include in subsequent rulemaking a mechanism to annually adjust the minimum study fee based on some type of inflation index.

Even if jurisdictional or technical discussions delay progress on all principles outlined in the ANOPR, FERC should prioritize developing rules related to this principle. In regions with a centrally cleared capacity market, RTO/ISOs establish demand curve construction methodologies and specific parameters to establish total MW quantity of reliability need for the market.³ Including large loads that pay study deposits and meet established readiness criteria in capacity market demand curves can be executed efficiently relative to other reforms. This reform alone would facilitate creation of forward market signals enabling timely investments necessary for reliable service to new large loads.

Principle 6: Shell Energy supports system protection facilities or system protection schemes as important tools to allow faster load interconnections without compromising the reliability of the BPS. This rulemaking should clarify the type of system protection facilities or system protection schemes envisioned, such as if the facilities will limit power flow during normal operations, or if the facilities only disconnect the co-located generation/load if an emergency drop is detected at the corresponding load/generation.

Shell Energy believes hybrid/co-located load and generation facilities should retain the right to request future study and full market participation without system protection facilities or system protection schemes limiting injection and withdrawal to low net MW values, where full grid access better suits their long-run operational needs, or after necessary network upgrades enter service. The necessity and feasibility of full-time hybrid/co-located operations without grid support may change over time as grid operators gain experience and tools to manage the risks of large loads, as new NERC standards mitigate some risks, as sufficient system-wide generation and transmission is built, and as long-range planning tools catch up to the necessary pace of load additions.

Principle 7: Shell Energy supports the principle that load curtailment can expedite studies. Load curtailment contributes to reliable operations during peak loads and is an important tool for

³ Guidance for how to LSEs and RTO/ISOs construct load forecasts for this purpose are prepared by regions with a centralized capacity market. See MISO BPM 11, PJM Manual 18, ISO-NE Manual 20, and NYISO Load Forecasting Manual for details on existing RTO/ISO roles in setting reserve or reliability margins, including provisions on adjusting reliability requirement for large loads.

cost-effective system planning. In rulemaking proceedings, FERC should examine how a hybrid/co-located load and generation resource model aligns with existing Demand Response (DR) markets and how to prevent the price suppression from demand response called through out-of-market mechanisms. Maintaining price and investment signals is especially critical with the potential scale of large load participation in demand response, which has historically only been 2-8% of regional peak loads.⁴

For example, does the ANOPR intend for the load to participate as price responsive or economic demand in energy market operations, or as “supply side” demand response that reduce capacity market demand curves, or both? RTO/ISO DR rules vary widely, so the payments to participating resources, penalties for non-response, measurement and verification requirements, market power mitigation, triggering criteria when an RTO/ISO is allowed to call on specific DR types, and eligibility for existing programs may create different and unexpected outcomes from region to region. Out-of-market load reduction DR is typically settled at zonal LMP, while in-market DR either through generation injection or load reduction is paid LMP at the Pnode. The rulemaking should detail what DR incentives or settlement mechanisms are appropriate for large loads- technology neutral solutions and affordability are important goals that should be accomplished without creating unintended price suppression or muting existing investment signals. Preferably, this should be executed solely as market-based demand response. If out-of-market deployments must occur, curtailment should take place immediately before firm load shedding, and steps should be taken to minimize any weakening of price signals resulting from these out-of-market actions.

Finally, if a hybrid/co-located load and generation resource has zero injection/withdrawal rights and is behind system protection facilities that restrict MW flows to or from the rest of grid, it is unclear how electrically isolated DR supports reliable bulk power system operations or cost-effective system planning.

C. Cost allocation

Principle 8: Shell Energy supports the principle that network upgrade costs are allocated according to established cost-causation principles. There are existing provisions in most regions to refund the financial security posted by generator interconnection customers to fund network upgrades over long time frames (through FTR credits, incremental ARR/FTR allocations or other crediting mechanisms). Analogous credits may be currently allocated to new network load customers (through incremental Auction Revenue Right (ARR) allocations) that fund network upgrades.

⁴ See NERC [2024 Long-Term Reliability Assessment Report](https://www.nerc.com/pa/RAPA/ra/Reliability%20Assessments%20DL/NERC_Long%20Term%20Reliability%20Assessment_2024.pdf) available at https://www.nerc.com/pa/RAPA/ra/Reliability%20Assessments%20DL/NERC_Long%20Term%20Reliability%20Assessment_2024.pdf

Large loads and hybrid/co-located load and generation customers covered by this ANOPR should qualify for either of these existing crediting mechanisms (as load or as generation) for network upgrades they fund; a sufficient record would need to be established on why large loads are differently situated enough to justify discriminatory treatment under tariff provisions applicable to other new generation and new load customers.

D. Non-discriminatory open access

Principle 9: Shell Energy supports study provisions that allow open access for entities other than incumbent transmission owners to construct necessary upgrades to interconnect load and generation. Many regions allow for generation interconnection customers to build their own transmission facilities before the incumbent transmission owner takes over transmission operations; no record has been established on why large loads that fund network upgrades should not be eligible for similar treatment.

E. Accounting of supply and demand, transmission rights, and ancillary services

Principle 10: Reply comments and subsequent rulemaking proceedings should clarify the interpretation and implementation of principle 10.

1. Shell Energy supports that existing generators are required to provide advance notice and be studied for the reliability impacts of new co-located loads.
2. Shell Energy does not support existing generation having discriminatory and punitive provisions that it must pay for network upgrades for engaging in a bilateral capacity sale. Existing generators that retire are not typically allocated these network upgrade costs, existing generators that exit a region to serve load in a different region are not allocated these network upgrade costs, and other generators selling power bilaterally are not allocated these network upgrade costs.
3. One of the clarifications needed under this principle is how generators, new or existing, that sell to load under a hybrid or co-located agreement are accounted for in centralized capacity markets or broader RA programs- if such sales are treated as retired resources, or if such sales are treated as traditional bilateral contracts. The proposal as written could imply that the generation component of hybrid/co-located load and generation resources does not have a must-offer obligation into capacity markets or may not be tracked under existing RA and procurement programs. This is not the current treatment of bilateral capacity sales under RA programs, and clarity is needed if the required system protection facilities under principle 6 merit different treatment and accounting under resource adequacy programs because their paired nature nets out the impact to capacity market clearing. Participation in centralized capacity markets or other RA programs is intertwined with generator outage reporting obligations, capacity interconnection rights or deliverability rights, day ahead energy must-offer obligations, and a number of other enforceable tariff provisions that ensure that grid operators have sufficient transparency into unit operations to maintain reliability. Shell Energy is

against implementation of principle 10 that reduces transparency or interpretability of basic market data.

4. Shell Energy supports accurate and transparent price signals in areas where grid impacts limit an existing generator's ability to sell to hybrid or co-located load. If it is the intention of principle 10 that grid operators have a right of first refusal to prevent generation resources from going behind system protection facilities to support co-located load, it is inconsistent with current practices and economic principles if such rights are exercised without providing adequate compensation for the resource. Similarly, shifting network upgrades costs to a market participant that will not benefit from the new infrastructure is inconsistent with cost causation principles.

5. Additionally, existing units likely have existing transmission access rights that could be released if they are instead serving on-site load. If a generator releases or sells NRIS, CIRs, or deliverability rights, existing generation queue customers can use that transmission capacity to enter operations faster instead of waiting for construction on new network upgrades to be completed. Existing generator interconnection study procedures have established process for allocating this "headroom" or surplus interconnection capacity to conventional generation queue customers.

6. Defining existing generation for the purposes of an SSR/RMR type studies requires additional record development. A resource may be included in forward capacity market or other resource adequacy procurement assumptions years in advance of entering commercial operations, and if a bilateral contract to serve co-located load removes them from existing RA supply/demand accounting frameworks then that could cause RA market volatility. Additionally, existing tariff provisions govern how a generating resource that has not yet entered commercial operations will still be responsible for network upgrades assigned to them in the interconnection process.

Shell Energy supports that principle 10 provides a transparency mechanism into bilateral transactions to serve large loads. Competitive markets require transparency so that shifts in underlying market fundamentals can be reflected in prices to signal and incentivize investment where needed. While quarterly EQR reporting may provide an existing and standardized transparency mechanism for intra-BAA capacity sales, the expedited study timelines proposed in the ANOPR may merit more frequent and timely ex-ante information on qualifying bilateral sales of capacity, instead of current ex-post EQR reporting.⁵

Principle 11: As discussed throughout these comments, using net withdrawal/ injection as the quantity of transmission service may be appropriate as an option for faster study, or interim operations, or to allow new large loads to be served by lower tier transmission service than NITS, but large loads and generation should retain the right to request and be studied for full service in the future.

⁵ See <https://www.ferc.gov/sites/default/files/2020-05/eqr-requirements-guide.pdf>

Principle 12: Shell Energy supports that ancillary services are assigned based on the load total, even for hybrid/co-located load and generation facilities. This proceeding should clarify if ancillary services include critical reliability services such as blackstart or region-specific products such as ramp capability.

Additional technical and engineering expertise should be solicited in this proceeding on how ancillary services can be served by the grid to a load resource isolated behind system protection facilities, and vice versa. If hybrid/co-located generation resources are committed to provide ancillary services capacity or follow an ancillary services dispatch in real time, how those services can reach and benefit the BPS from a generator without energy injection rights should be confirmed, especially if a penalty structure on energy injection as described in principle 6 is established.

F. Administration and implementation

Principle 13: Loads that have entered study under existing mechanisms should not be delayed while a potential new rulemaking unfolds, or subject to additional uncertainty that existing agreements could be invalidated by this rulemaking. It is nonsensical to delay existing and ongoing stakeholder processes for regional reforms that will speed load interconnection while nationwide policy tradeoffs are debated in this rulemaking; as with all rulemakings the final compliance filings will contain regional variation or require incremental tariff revisions to conform.

Existing loads under study, and ongoing stakeholder processes to develop expedited large load processes should prioritize how to improve transparency to the market about the volume of load under study and readiness status of the load. Financial deposits to continue to progress in the load queue will ensure that load currently under study is validated and non-duplicative, and will improve reliability and confidence in load forecasts for markets to meet that demand.

It is important that FERC and the RTO/ISOs assess the impact of these expedited requests on existing requests in current generator interconnection queues. Order No. 2023 sought to reduce the risk of last-minute withdrawals, dramatic swings in allocated network upgrade costs, and cascading restudies because of unreliable network models. Expedited study processes envisioned under this ANOPR should not cause delays, uncertainty, or allocation of new costs to generator interconnection customers already in queues.

Some principles laid out in the ANOPR may overlap with Order No. 1920 compliance filings, which will not complete staggered compliance until the middle of 2027.⁶ If and how large loads interconnected under expedited procedures must be included in long-term load scenarios or forecasts, or the ex-ante plan for how the costs of long-term transmission identified under Order No. 1920 procedures will be allocated, and identifying transmission facilities for right sizing are

⁶ <https://www.ferc.gov/news-events/news/order-no-1920-compliance-filings-schedule>

potential areas of overlap. The Commission should address how they will assess Order No. 1920 compliance filings that include treatment of large loads.

Principle 14: NERC has been actively working through their Large Load Task Force to identify the risks to the bulk power grid from large loads and identify policy recommendations. The task force did not have hybrid/co-located generation operations identified as priority items, but the policy recommendations expected to be released Quarter 1 2026, the draft reliability guidelines scheduled for NERC approval May 2026, and the NERC Data Center Technical Conference scheduled for February 2026 will provide important insights to the technology and engineering underpinning reliable integration of large loads to the bulk power system.⁷

III. Conclusion

Shell Energy applauds the FERC for recognizing the seriousness and urgency of reforms needed to quickly interconnect large loads. The proposals highlight opportunities to standardize processes, increase transparency so competitive markets can respond with solutions, and meaningfully speed up the study timelines for new large loads without inappropriately shifting the cost to existing ratepayers, impacting established generator interconnection processes, or risking reliable operations. Shell Energy appreciates the opportunity to comment on the specific implementation details that FERC should consider in this rulemaking.

Respectfully Submitted,

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⁷ See https://www.nerc.com/globalassets/who-we-are/standing-committees/rstc/lltf/lltf_presentations_september_25_2025.pdf

CERTIFICATE OF SERVICE

I hereby certify that I have served the foregoing document upon each person listed on the official service list compiled by the Secretary in this proceeding.

Dated at Washington, D.C., this 21st day of November, 2025.

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